

1. System Overview

1.1 Purpose and Scope The Hotel Booking Management System is a web-based application designed to facilitate the reservation process for guests and streamline daily operations for hotel staff. The system aims to replace manual booking methods with an automated platform that handles room availability, guest registration, secure payments, and administrative reporting.

1.2 Key Stakeholders and User Types

- **Guest:** An external user who searches for rooms, creates an account, and makes bookings.
- **Receptionist:** A staff member who manages guest check-ins/check-outs and updates room status.
- **Administrator:** A super-user who manages the entire system, including user accounts, room inventory, and financial reports.

1.3 Major System Modules

- **User Authentication Module:** Registration, login, and password management.
- **Booking Engine:** Search functionality, availability checking, and reservation creation.
- **Room Management Module:** CRUD (Create, Read, Update, Delete) operations for room details.
- **Reporting Module:** Generation of revenue and occupancy reports.

2. Functional Requirements

These are the 10 required features, written in the "The system shall..." format.

- **FR-01:** The system shall allow guests to register for an account by providing their name, email address, and phone number.
- **FR-02:** The system shall allow registered guests to search for rooms by entering check-in/check-out dates and selecting a room type (e.g., Deluxe, Standard).
- **FR-03:** The system shall display detailed information for each room, including photos, amenities, price per night, and current availability status.
- **FR-04:** The system shall prevent double-booking by automatically validating room availability before confirming any reservation transaction.
- **FR-05:** The system shall allow guests to cancel their reservation at least 24 hours prior to the check-in date.
- **FR-06:** The system shall generate an automated email notification to the guest immediately after a successful booking or cancellation.
- **FR-07:** The system shall allow the Administrator to add, update, and delete room inventory details, including pricing and images.
- **FR-08:** The system shall allow the Receptionist to update the status of a room (e.g., "Available", "Occupied", "Under Maintenance") in real-time.
- **FR-09:** The system shall securely process guest payments and generate a digital invoice/receipt upon completion.
- **FR-10:** The system shall allow the Administrator to generate monthly performance reports showing total revenue and room occupancy rates.

3. Non-Functional Requirements

These define how well the system performs.

- **NFR-01 (Performance):** The system shall load search results within 2 seconds to ensure a smooth user experience.
- **NFR-02 (Security):** All sensitive user data, including passwords and credit card information, shall be encrypted using industry-standard protocols (e.g., AES-256).
- **NFR-03 (Availability):** The system shall maintain 99.9% uptime to allow for 24/7 booking access by guests.
- **NFR-04 (Scalability):** The system shall support up to 1,000 concurrent users without performance degradation during peak holiday seasons.
- **NFR-05 (Usability):** The system interface shall be responsive and fully functional on both desktop and mobile devices.